



sticking together

The glue force that holds quarks together inside a nucleon can leak out of that nucleon and affect quarks in the nucleon next door. It is this leakage of the glue force that we detect as the strong force that holds atomic nuclei together.

tracking down the boson

The British physicist, Peter Higgs, developed a model to explain why particles have mass, this predicts the existence of a kind of particle—now called the Higgs boson—but scientists are still searching for the Higgs particle.

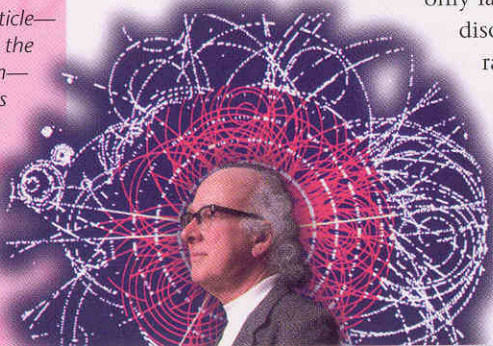
This whole package of ideas is closely modeled on quantum electrodynamics, and is known as quantum chromodynamics, or QCD. The glue force between quarks leaks out from one nucleon to affect its neighbors relatively slightly, and provides the strong force that holds the nucleus together. The real color force is a lot stronger than the “strong” nuclear force, which overwhelms the electrical repulsion of protons on the nuclear scale.

nuclear decay

There is just one other interaction that affects the behavior of particles in the quantum world, and the way it works highlights why physicists prefer the term “interaction” to the term “force.” The weak interaction, as it is known, also operates only on the scale of nuclei and nucleons. Its most important property is that it is responsible for some of the processes involved in radioactive decay and this can be seen best from an example—the process known as beta decay, in which a neutron is transformed into a proton.

The process derives its name from when this kind of radioactivity was first discovered in the closing years of the 1800s. Then, the radiation produced was called beta radiation (another kind of radiation had already been dubbed alpha radiation); it was only later that it was discovered that “beta rays” are in fact a stream of fast-moving electrons.

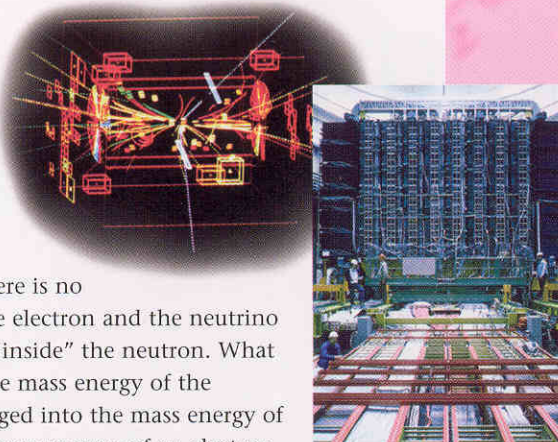
In beta decay, a neutron (either sitting on its own in space, or



inside a nucleus) transforms itself into a proton by spitting out an electron and a particle known as a neutrino. It is important to

appreciate that there is no sense in which the electron and the neutrino were at any time “inside” the neutron. What happens is that the mass energy of the neutron is rearranged into the mass energy of a proton plus the mass energy of an electron and a neutrino. The neutron does this by ejecting (from one of its constituent quarks) the particle that is the “force carrier” of the weak interaction, analogous to the photon in QED and the gluon in QCD. But this particle, which is somewhat boringly called an intermediate vector boson, carries one unit of negative electric charge. The energy of the boson is then rapidly converted into the mass energy of a neutrino and an electron. The electron has the unit of negative charge, so to balance the books, the neutron left behind now has one unit of positive charge and has become a proton. There is also a similar boson, which carries one unit of positive charge, and one which, like the photon, has no charge at all; these are involved in other kinds of weak interaction.

The discovery of all three of these bosons in experiments at CERN in 1983, following the triumph of QCD in the 1970s, put the seal on what has become the standard model of the quantum world. Everything on Earth can be explained in terms of quarks, electrons and their associated neutrinos, and four interactions—gravity, electromagnetism, the weak nuclear interaction, and the glue force underpinning the strong nuclear interaction.



picturing particles

The UA1 detector at CERN, (pictured above during its construction in 1981), is the size of a house. When beams of subatomic particles collide in the center of the detector, they produce showers of other particles, which are tracked by the UA1 assembly. A computer then produces a picture of the particle shower. The image (above left) records the first detection of a Z particle, one of the intermediate vector bosons, in 1983.